

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4  #include "commonType.h"
5  #include "initAndTerm.h"
6  #include "dataControl.h"
7
8  #define TOT_MENU_CNT 5
9  int vertexCreatedCnt = 0, vertexCnt = 0;
10 extern LQType *Q;
11
12 int getSelection(void){
13     int selectNum = 0;
14
15     printf("\n");
16     printf("1. input total vertex count\n");
17     printf("2. input vertex and that of near vertex\n");
18     printf("3. delete near vertex of vertex\n");
19     printf("4. print vertex and bfs result\n");
20     printf("5. end\n");
21     printf("select menu: ");
22     scanf("%d", &selectNum);
23     while(getchar() != '\n');
24
25     return selectNum;
26 }
27 int exeFunc(void (**pF)(graphType**), int sel, graphType** graph, int *ret){
28     pF[sel](graph);
29     *ret = 0;
30     return 0;
31 }
32 void selectInputVertexCnt(graphType **graph) {
33     int i = 0;
34
35     printf(" current Vertex Count : %d\n", vertexCreatedCnt);
36     insertTotalVertex(&vertexCreatedCnt);
37     createBFSGraph(*graph, vertexCreatedCnt);
38
39     for (i = 0; i < vertexCreatedCnt; i++) {
40         insertEachVertex(*graph, i, vertexCreatedCnt);
41     }
42 }
43 void selectInputNearListOfVertex(graphType **graph){
44     int *tempVL;
45     int i = 0;
46     vertexCnt = 0;
47
48     if ((*graph)->n == 0) {
49         printf(" !!!there is no vertex count\n");
50         return;
51     }
52     createVertexList(&tempVL, vertexCreatedCnt);
53     vertexCnt = requestTargetVertex(1);
54     if ((*graph)->nearListHead[vertexCnt] != NULL) {
55         eraseNearListOfVertex(*graph, vertexCnt);
56     }
57     inputVertexList(&tempVL, vertexCreatedCnt, vertexCnt);
58     sortNearList(&tempVL, vertexCreatedCnt);
59     for(i=0; i < vertexCreatedCnt; i++){
60         if(tempVL[i] != -1){
61             insertEachEdge(*graph, vertexCnt, tempVL[i]);
62         }
63     }
64     releaseVertexList(&tempVL);
65 }
66 void selectDeleteNearListOfVertex(graphType **graph) {
67     int i = 0;
68     vertexCnt = 0;
69
70     if ((*graph)->n == 0) {
71         printf(" !!!there is no vertex count\n");
72         return;
73     }
74     vertexCnt = requestTargetVertex(2);
75     deleteNearList(*graph, vertexCnt);
76 }
77 void selectPrint(graphType **graph) {
78     vertexCnt = 0;
79
80     if ((*graph)->n == 0) {
81         printf(" !!!there is no vertex count\n");
82         return;
83     }
84     printNearList(*graph);
85     while (vertexCnt < vertexCreatedCnt) {
86         checkExistAcrossEdge(*graph, vertexCnt, vertexCreatedCnt);
87         vertexCnt++;
88     }
89     bfsNearList(*graph, 0);
90 }
91 void selectEnd(graphType **graph){
92     releaseNearListHead(*graph, vertexCreatedCnt);
93     releaseBFSGraph(*graph);
94     free(*graph);
95 }
96 int main(int argc, char *argv[]){
97     graphType *graph;
98     int selNum = 0, res = -1;
99
100     graph = (graphType*)malloc(sizeof(graphType));
101     memset(graph, '\0', sizeof(graph));
102     void (*pF[TOT_MENU_CNT])(graphType**) = {selectInputVertexCnt
103         , selectInputNearListOfVertex, selectDeleteNearListOfVertex, selectPrint, selectEnd};
104     while(selNum != TOT_MENU_CNT){
105         selNum = getSelection();
106         if(selNum > 0 && selNum < (TOT_MENU_CNT+1) ){
107             exeFunc( pF, (selNum-1), &graph, &res );
108             printf(" execute result: %d\n", res);
109         }else{
110             printf(" !!!select menu again\n");
111             continue;
112         }
113     }
114
115     return 0;
116 }
117
```